WEEK -5

Minor Project (BTCS607N)-Weekly Progress Report

Project Title: Drone-Based Intelligent System for Apple Orchard Management in Himachal Pradesh.

Group ID: G9

Branch & Section: BTech CSE (AIML with Microsoft), AIDS -A

Date: 20-02-2026

1. Executive Summary

A. **Current Sprint:** Sprint 1 (Issue – 5)

B. **Sprint / Issue Goal:** Complete Architecture & Module Design, create a detailed Sequence Diagram to define system interactions, and collect the dataset for the project.

C. **Status:** On Track

2. UML & OOAD Progress

A. **Diagrams Completed this week:**

1. **UML Sequence Diagram**: Defined the interaction flow between the system components (User, Web Interface, Disease Detection, etc.) for detecting apple diseases and estimating yield.

B. **Diagrams Revised:**

1. Ensured the proper sequencing of events, starting from image upload to health classification, disease detection, and advisory.

C. **OO Concept Applied:**

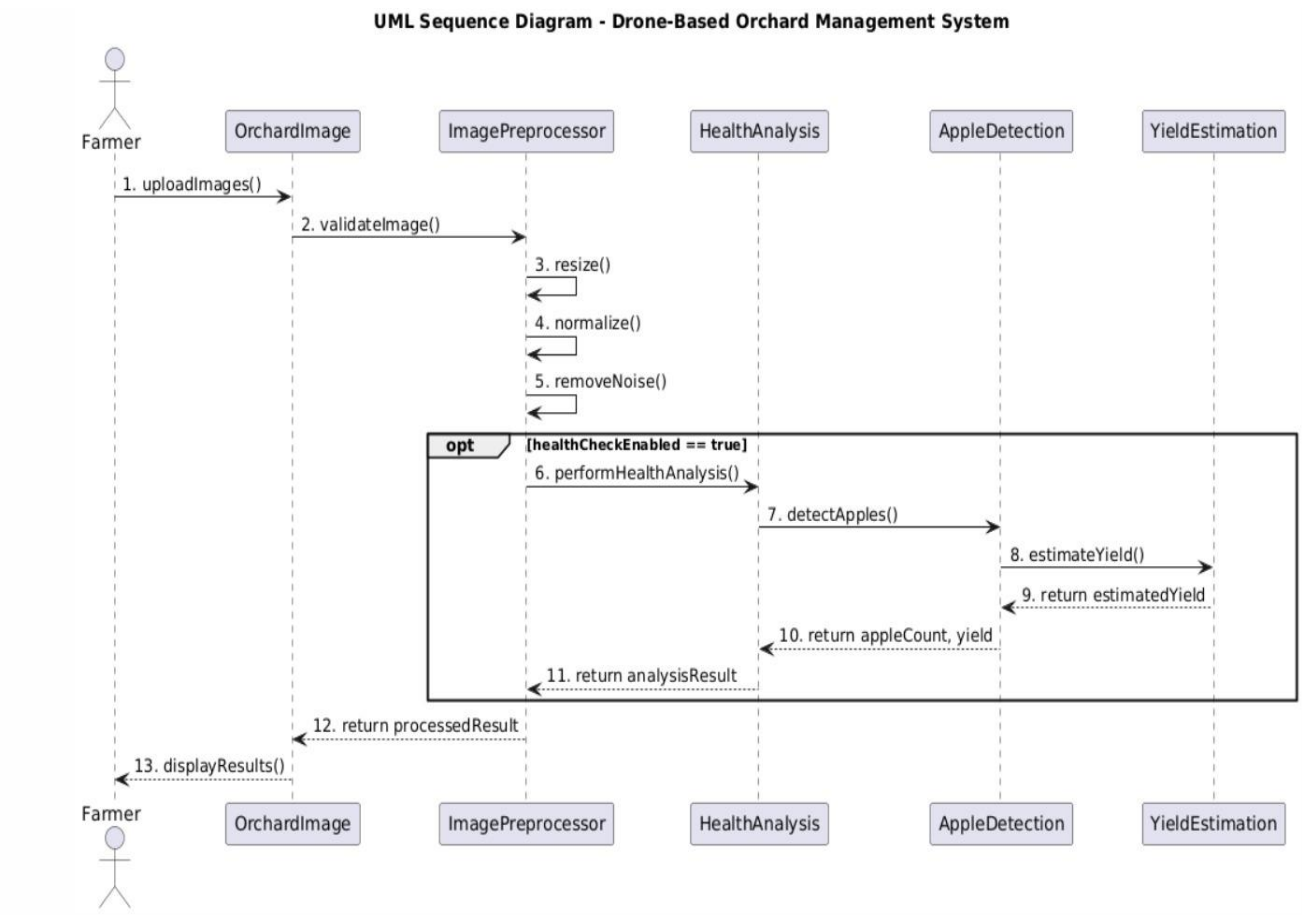
1. **Message Passing**: Described the sequence of messages exchanged between different system components.
2. **Parallel Processing**: Illustrated parallelism for disease detection and tree health analysis.
3. **Exception Handling**: Incorporated error handling for invalid image uploads or failed analyses.

Main Workflow of the System (Sequence of Interactions):

1. User uploads orchard image through the Web Interface.
2. Image is sent to the Drone Image Simulation Module.
3. The image is processed by the Disease Detection System.

4. The results from disease detection are forwarded to the Tree Health Classification System.
5. The classification results are used by the Advisory System for yield prediction and suggestions.
6. Final results are displayed to the User**.**

Sequence Diagram:



3. Dataset Collection

Datasets Collected for Apple Disease Detection:

- The dataset collection was completed for training the disease detection system, covering diseases such as **Apple Scab**, **Powdery Mildew**, and **Cedar Apple Rust, Brown Rot**.
- Links to the collected datasets:

1. **Apple Scab (Venturia inaequalis)**: https://www.kaggle.com/code/shamikrana/apple-leaves-disease-prediction/input?select=APPLE_DISEASE_DATASET
2. **Powdery Mildew (Podosphaera leucotricha)**: https://www.kaggle.com/code/yashvi/classify-diseases-in-apple-trees-beginner/input?select=test_images
3. **Cedar Apple Rust (Gymnosporangium spp.)**:

   https://www.kaggle.com/code/prakharsinghchouhan/getting-started-with-vision-transformers/input

   https://www.kaggle.com/code/nirmalgaud/apple-leaf-disease-detection/input
4. **Brown Rot (Monilinia spp.)**: https://www.kaggle.com/code/faizurazadri/apple-leaf-disease-classification-using-vit/input

4. Technical Execution (DevOps & Git)

A. **Jira Tasks Completed:**

   1. Designed and validated the UML Sequence Diagram.
   2. Completed dataset collection for apple diseases.

Link://ayushshrivastava.atlassian.net/jira/software/projects/DBISFAOM/boards/2?atlOrigin=eyJpIjoiY2I0Y2FjMmQ2MTliNDBlM2FjYzEyN2U1MjdkY2QzNGMiLCJwIjoiaiJ9

B. **Git Activity:**

   1. Active branches: main
   2. Documentation updated with Sequence Diagram and dataset information.

Link : https://github.com/yp1818/Drone-Based-Apple-Orchard-Management

C. **CI/CD Status:** Not applicable (design phase – no code yet)

5. Testing Summary

A. **Total Unit Tests:** Not applicable in design phase

B. **Pass Rate:** Not applicable in design phase

C. **Key Bug Identified:** No major issues were identified during this phase.

6. Goals for Next Week

a. **Detailed Class Design & Packaging**: Apply **SOLID principles** and **Design Patterns**.
   UML Deliverables:
   1. **Class Diagram**: Methods, visibility, interfaces.
   2. **Package Diagram**: Organize classes into logical modules.
b. **Continue Advisory System & Yield Estimation Development.**
c. **Integrate Disease Detection with Frontend** for real-time analysis.

Mentor Signature **Project Guide Signature**